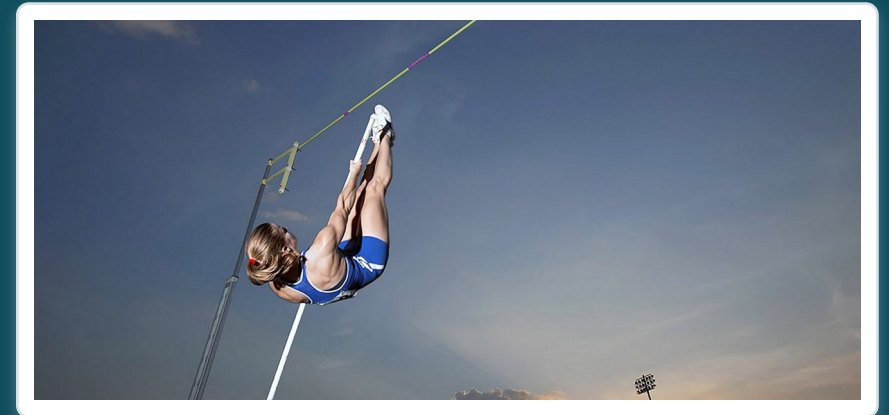


CHAPTER 24

Metabolism and Nutrition

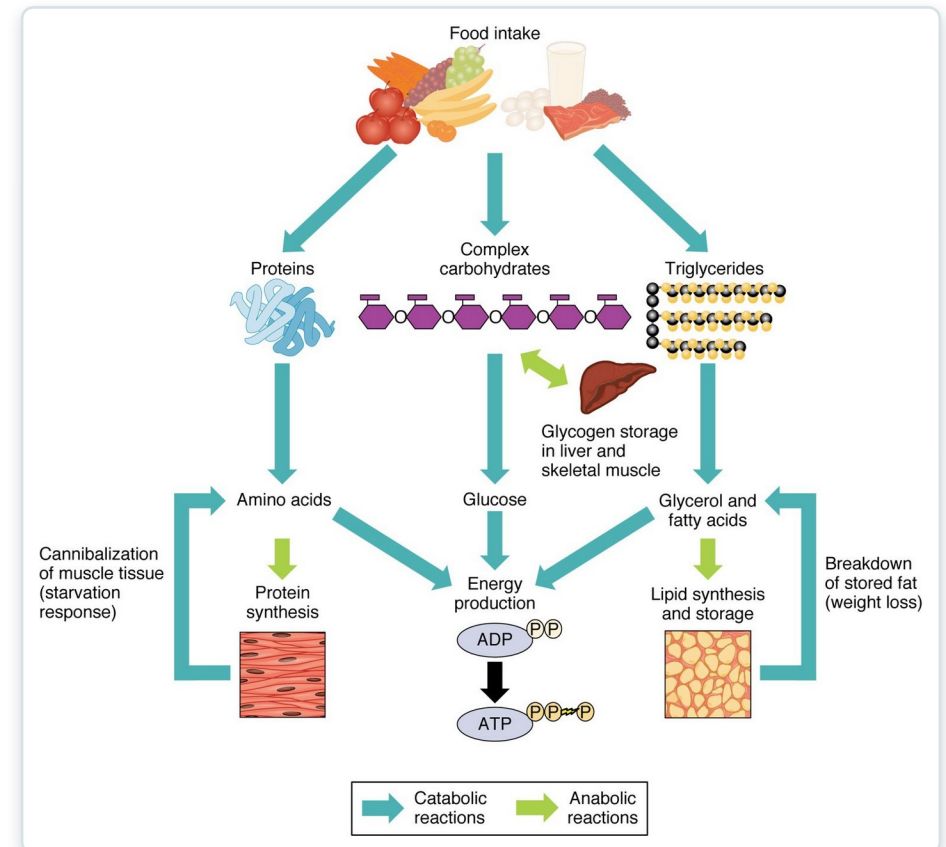
How the body turns food into energy and building blocks.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the processes involved in anabolic and catabolic reactions
- 2 List and describe the steps necessary for carbohydrate, lipid, and protein metabolism
- 3 Explain the processes that regulate glucose levels during the absorptive and postabsorptive states
- 4 Explain how metabolism is essential to maintaining body temperature (thermoregulation)
- 5 Summarize the importance of vitamins and minerals in the diet Eating is essential to life



Metabolism: building up and breaking down.



24.1

SECTION

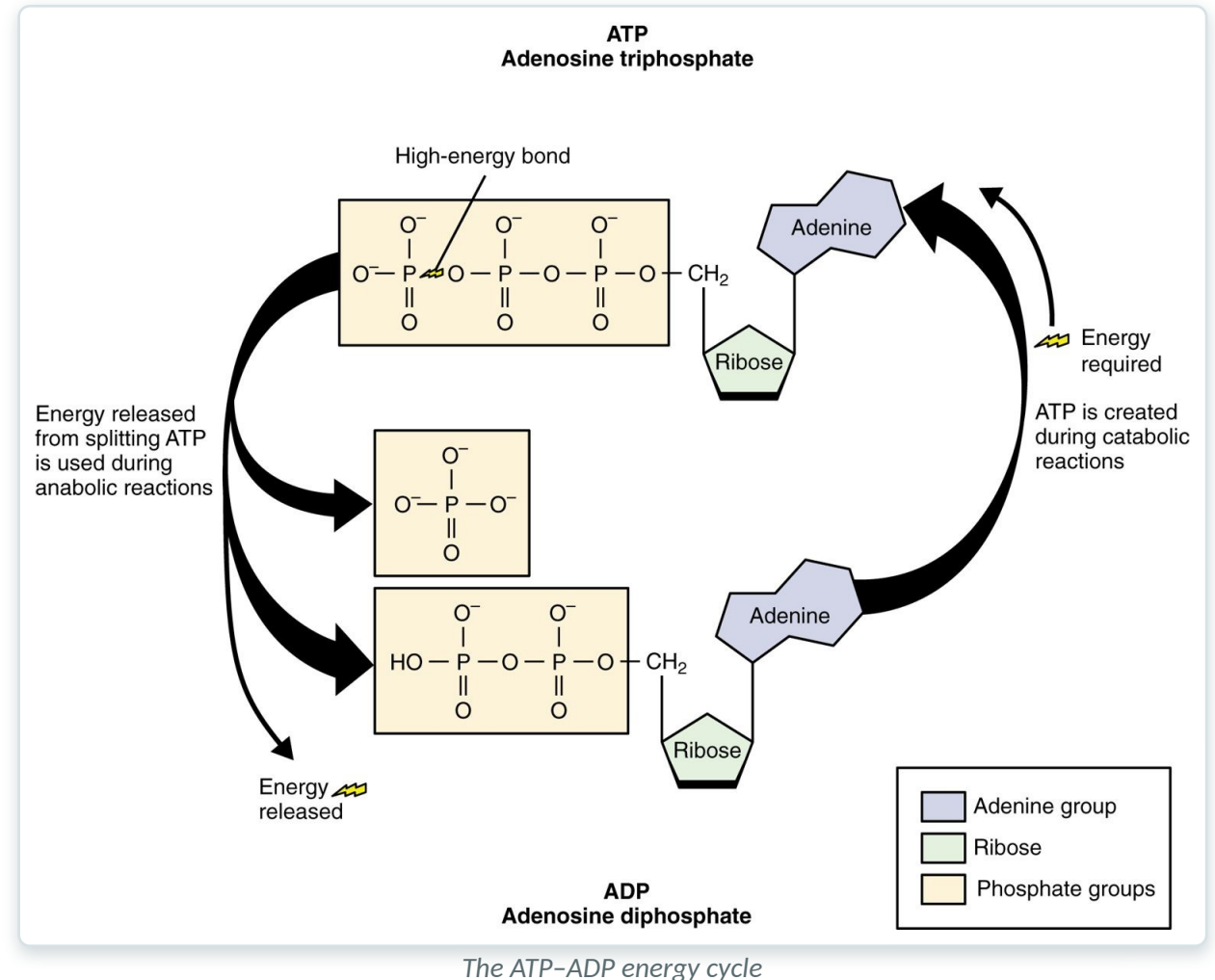
Overview of Metabolic Reactions

Metabolism — the chemical reactions that power the body.

SECTION 24.1

ATP — The Energy Currency

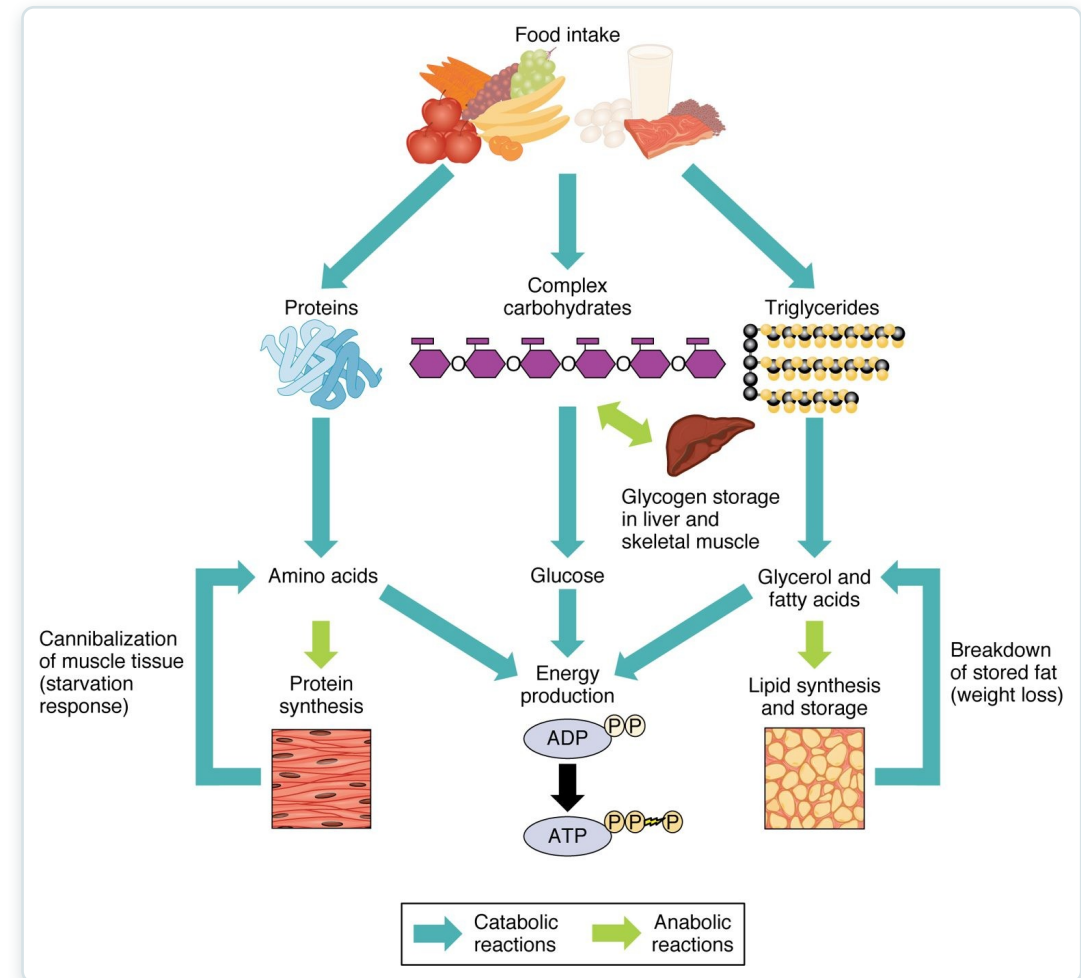
- ATP stores and delivers energy
- Made by breaking down nutrients
- Releases energy when split to ADP
- Powers nearly all cell work



SECTION 24.1

Anabolism & Catabolism

- Metabolism = all chemical reactions
- Catabolism breaks molecules down
- Anabolism builds molecules up
- Food fuels both



Catabolic and anabolic reactions



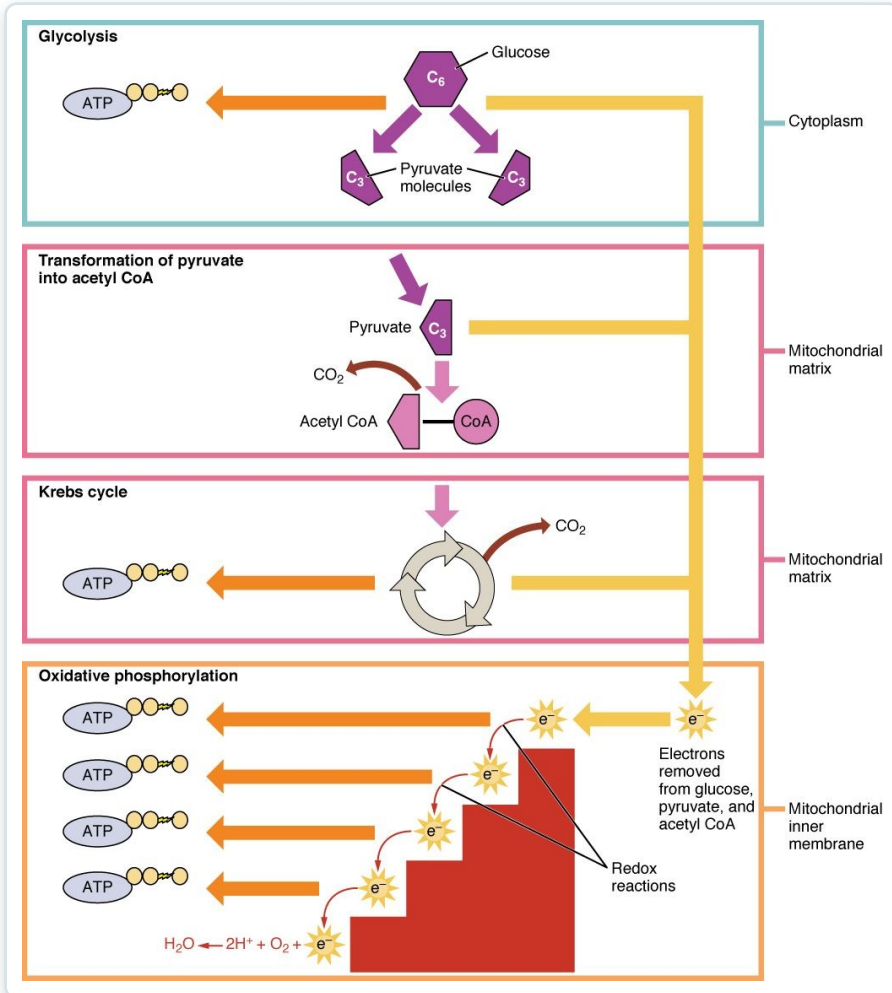
24.2

SECTION

Carbohydrate Metabolism

How the body extracts energy from carbohydrates.

Cellular Respiration



An overview of cellular respiration

- Glucose is broken down for ATP
- Glycolysis splits glucose
- The Krebs cycle extracts energy
- The electron transport chain makes most ATP



24.3

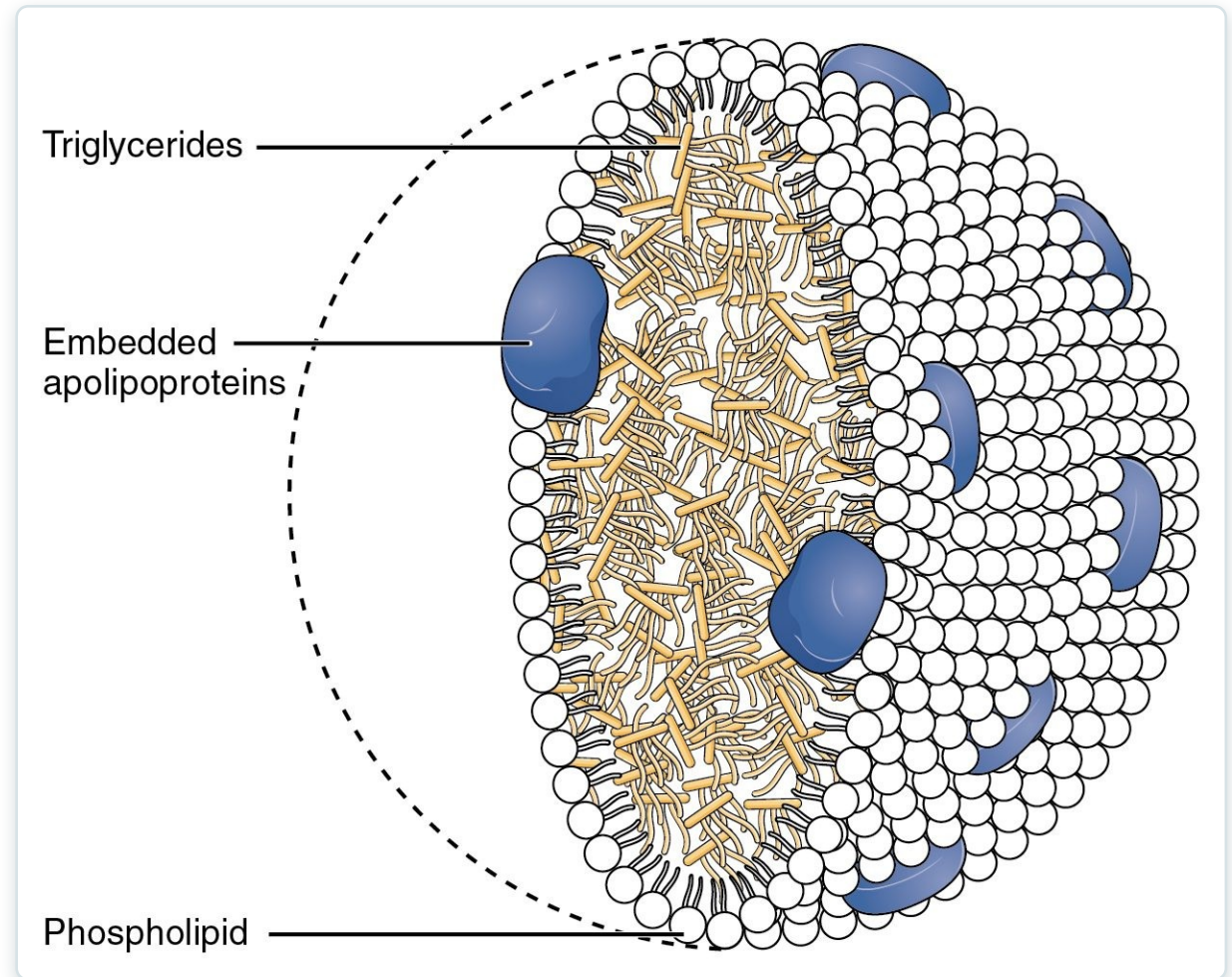
SECTION

Lipid Metabolism

How the body uses and transports fats.

Lipoproteins

- Fats don't dissolve in blood
- Carried in lipoprotein packages
- LDL delivers cholesterol to tissues
- HDL returns it to the liver



Lipoprotein structure



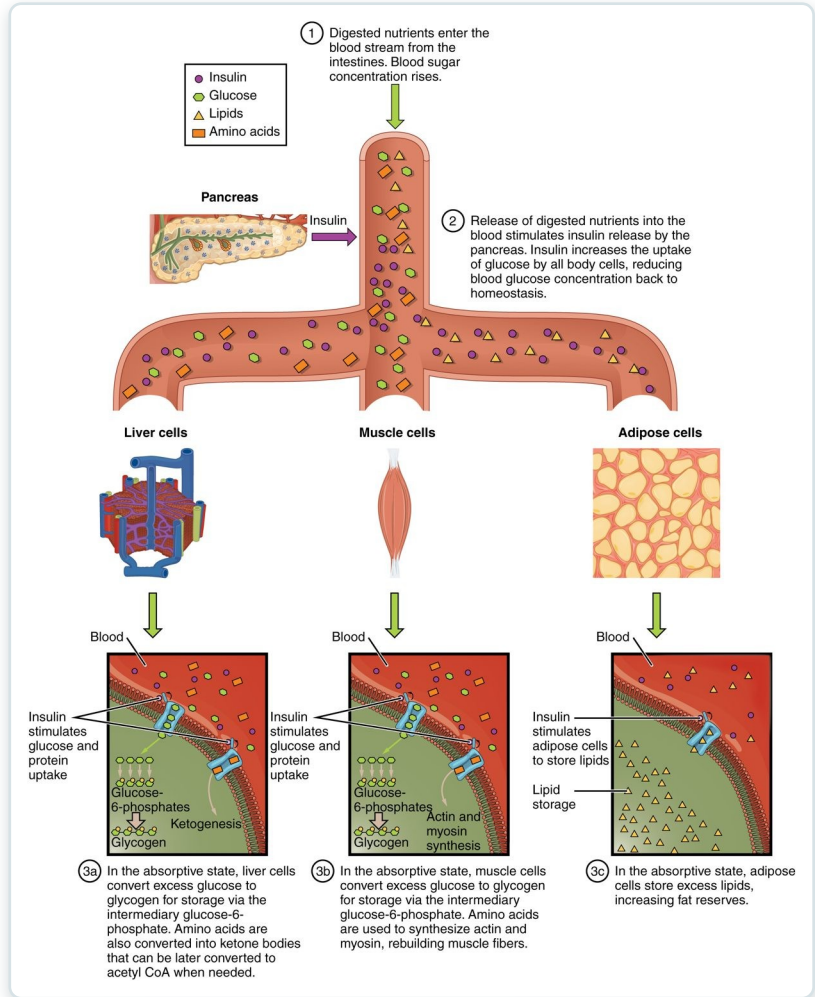
24.5

SECTION

Metabolic States of the Body

How metabolism shifts between the fed and fasting states.

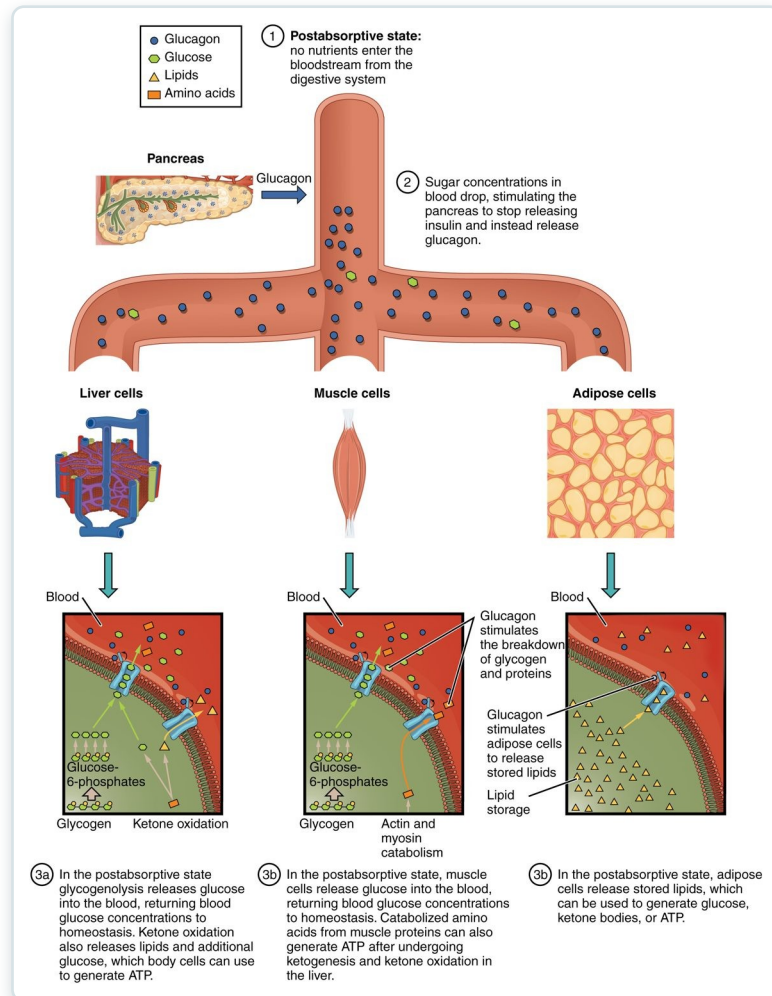
The Absorptive State



The absorptive (fed) state

- The "fed" state after a meal
- Blood glucose is high
- Insulin drives storage
- Glucose, fat, and protein are stored

The Postabsorptive State



The postabsorptive (fasting) state

- The fasting state between meals
- Blood glucose tends to fall
- Glucagon mobilizes stores
- Glycogen and fat are broken down



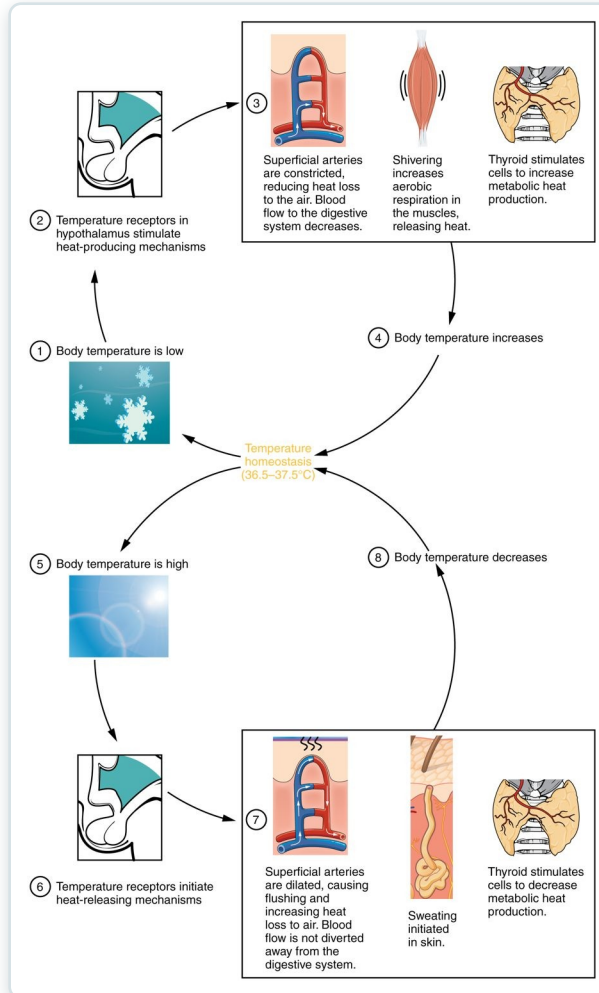
24.6

SECTION

Energy and Heat Balance

How the body balances energy and regulates temperature.

Heat Balance & Thermoregulation



Heat balance and thermoregulation

- Metabolism produces heat
- The hypothalamus is the thermostat
- Sweating and blood flow lose heat
- Shivering generates heat



24.7

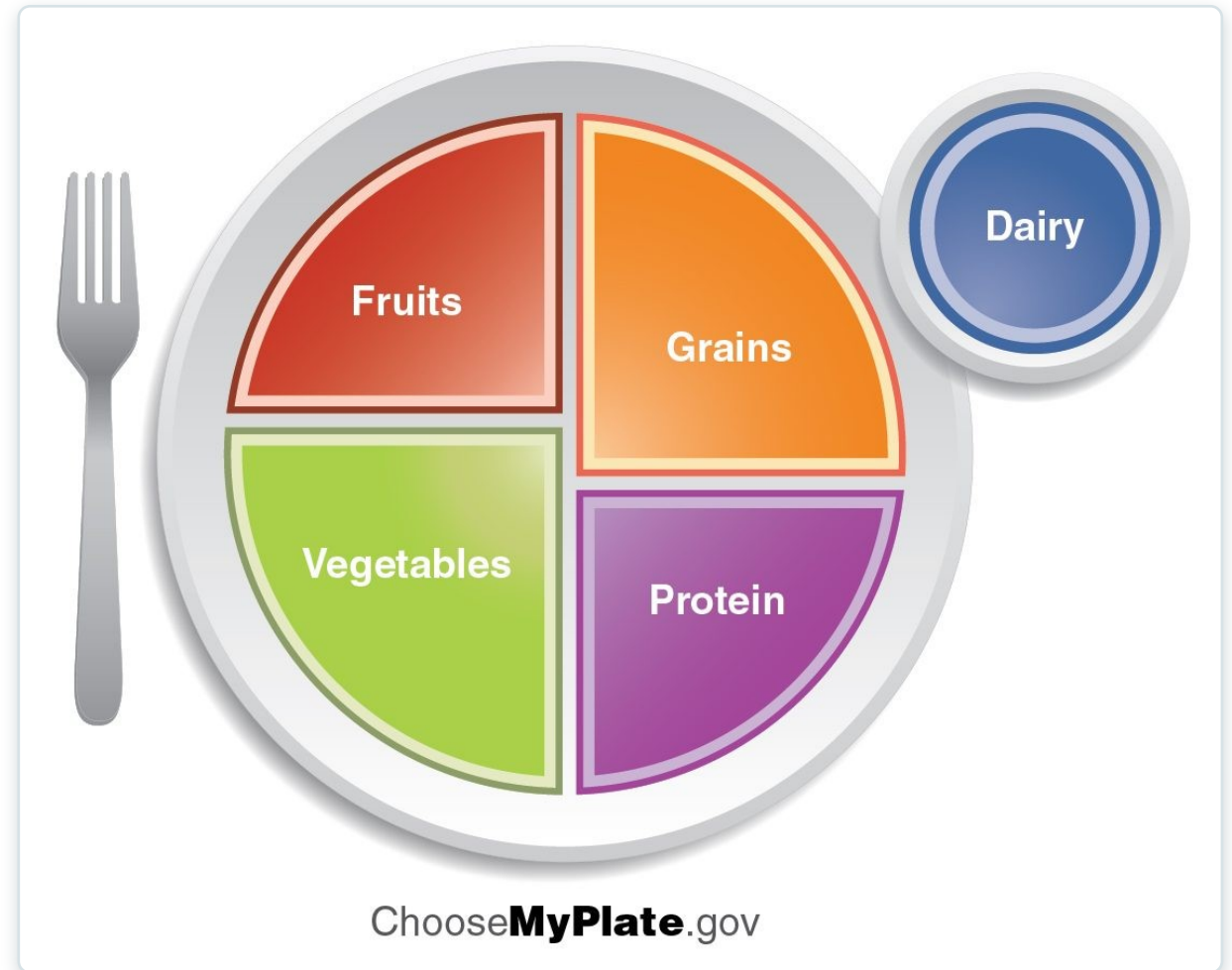
SECTION

Nutrition and Diet

The nutrients the body needs and a balanced diet.

Nutrition & MyPlate

- A balanced diet supplies all nutrients
- Carbohydrates, fats, proteins for energy
- Vitamins and minerals in small amounts
- MyPlate guides healthy portions



MyPlate dietary guidelines

Key Takeaways

- Metabolism is all the body's chemical reactions — catabolism releasing energy, anabolism using it.
- ATP is the energy currency that powers cellular work.
- Cells extract energy from glucose through cellular respiration.
- Metabolism shifts between storing fuel after meals and mobilizing it while fasting.
- The body balances energy intake, heat, and nutrient needs through diet.

CLINICAL CONNECTIONS

Why Metabolism & Nutrition Matter in Practical Nursing

Diabetes

Disrupted glucose metabolism affects energy and many organs.

Cholesterol

The LDL and HDL balance shapes cardiovascular risk.

Nutrition Teaching

Balanced-diet guidance supports patient health.

Fever

A raised hypothalamic set point during illness.

Malnutrition

Nutrient deficiencies impair healing and immunity.

Obesity

Chronic energy surplus stored as fat raises health risks.